

torch.combinations#

<https://pytorch.org/docs/stable/generated/torch.combinations.html>

Description:

Compute combinations of length `rrr` of the given tensor. The behavior is similar to python's `itertools.combinations` when `with_replacement` is set to `False`, and `itertools.combinations_with_replacement` when `with_replacement` is set to `True`. input (Tensor) – 1D vector. `r` (int, optional) – number of elements to combine `with_replacement` (bool, optional) – whether to allow duplication in combination A tensor equivalent to converting all the input tensors into lists, do `itertools.combinations` or `itertools.combinations_with_replacement` on these lists, and finally convert the resulting list into tensor.

Function Signature

```
torch.combinations(input: Tensor, r: int = 2,  
with_replacement: bool = False) → seq#
```

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> a = [1, 2, 3]
>>> list(itertools.combinations(a, r=2))
[(1, 2), (1, 3), (2, 3)]
>>> list(itertools.combinations(a, r=3))
[(1, 2, 3)]
>>> list(itertools.combinations_with_replacement(a, r=2))
[(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)]
>>> tensor_a = torch.tensor(a)
>>> torch.combinations(tensor_a)
tensor([[1, 2],
        [1, 3],
        [2, 3]])
>>> torch.combinations(tensor_a, r=3)
tensor([[1, 2, 3]])
>>> torch.combinations(tensor_a, with_replacement=True)
tensor([[1, 1],
        [1, 2],
        [1, 3],
        [2, 2],
        [2, 3],
        [3, 3]])
```

Detailed Documentation

Documentation

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`input` (Tensor) – 1D vector.

`r` (int, optional) – number of elements to combine

`with_replacement` (bool, optional) – whether to allow duplication in combination

A tensor equivalent to converting all the input tensors into lists, do `itertools.combinations` or `itertools.combinations_with_replacement` on these lists, and finally convert the resulting list into tensor.